

Spectral Collapse Without Collapse: SVD Dynamics of LoRA Adapters for Low-Resource Turkic Languages

Anonymous ARR Submission

Abstract

Spectral collapse—the concentration of adapter weight matrices into a single dominant singular direction—has been proposed as a diagnostic for LoRA training failure. We systematically test this hypothesis by fine-tuning Gemma-2-9B with LoRA on three low-resource Turkic languages (Kyrgyz, Kazakh, Uzbek) across six experimental configurations varying rank ($r \in \{16, 64\}$), learning rate ($2 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}$), training duration (3–10 epochs), and cross-lingual transfer (Kazakh $\rightarrow$ Kyrgyz).

We report three main findings. First, we **refute the spectral energy threshold hypothesis**: no experiment reaches $SE > 0.59$ (well below the proposed 0.7 threshold), yet configurations with $r = 64$ suffer complete functional collapse (NER F1 = 0 across all languages, cross-lingual perplexity > 600). Spectral energy actually *decreases* during training as effective rank grows. Second, we identify **Frobenius norm growth** as a more reliable indicator of functional degradation in our setting (Gemma-2-9B, 4-bit quantization): pathological configurations exhibit 15–38 $\times$ norm growth versus only +13% for the best-performing transfer setup. Third, we demonstrate that **sequential Kazakh $\rightarrow$ Kyrgyz transfer** yields +61% NER F1 improvement (0.253 vs. 0.157) while retaining Kazakh knowledge (perplexity 23.5 vs. 47.6 for direct Kyrgyz training), suggesting that related-language transfer is an effective strategy for extremely low-resource Turkic varieties.

We additionally characterize severe subword fragmentation ($3.2\times$ vs. English) as a compounding factor for all three languages and release our SVD monitoring toolkit for reproducible spectral analysis of LoRA training dynamics.

1 Introduction

Low-Rank Adaptation (Hu et al., 2022) has become the dominant approach for adapting large

language models (LLMs) to new domains and languages, particularly when computational resources are limited. By decomposing weight updates into low-rank matrices $\Delta W = BA$, LoRA achieves parameter efficiency while preserving most of the pretrained model’s capabilities. However, the internal dynamics of LoRA weight matrices during training—specifically their spectral properties—remain poorly understood, especially for morphologically complex, low-resource languages.

Recent work has identified *spectral collapse* as a potential failure mode of LoRA fine-tuning, where adapter weight matrices degenerate into rank-one approximations dominated by a single singular value (Biderman et al., 2024; Jang et al., 2024). Biderman et al. (2024) proposed that when the spectral energy (SE)—the ratio of the largest singular value to the sum of all singular values—exceeds a threshold of approximately 0.7, the adapter has effectively collapsed and further training becomes unproductive. However, these findings were established primarily on English-language tasks with relatively large training corpora.

Turkic languages present a particularly challenging testbed for LoRA adaptation. Kyrgyz, Kazakh, and Uzbek are agglutinative languages with rich morphological systems, employ both Cyrillic and Latin scripts, suffer from severe underrepresentation in multilingual LLM pretraining data, and exhibit extreme subword fragmentation under standard tokenizers designed for high-resource languages. These properties make Turkic language adaptation a stress test for parameter-efficient fine-tuning methods.

Turkic languages are particularly informative for this investigation due to their extreme subword fragmentation under standard tokenizers ($3.2\times$ more tokens per word than English), which forces LoRA adapters to make larger weight modifica-

tions to accommodate poorly-covered morphology.

In this work, we systematically investigate the spectral dynamics of LoRA adapters trained on three Turkic languages, testing two hypotheses:

- **H1 (Spectral Energy Threshold):** Spectral energy exceeding 0.7 predicts functional degradation of LoRA adapters (Biderman et al., 2024).
- **H2 (Cross-Lingual Transfer):** Initializing LoRA adapters from a related-language checkpoint (Kazakh $\rightarrow$ Kyrgyz) slows spectral degradation and improves downstream task performance.

Our contributions are as follows:

1. We **refute the SE threshold hypothesis** for low-resource LoRA fine-tuning: maximum SE never exceeds 0.59, yet functional collapse occurs through alternative mechanisms (Section 4.3).
2. We propose **Frobenius norm growth ratio** as a scale-aware alternative diagnostic that reliably separates healthy from pathological training regimes (Section 4.4).
3. We demonstrate the **effectiveness of cross-lingual transfer** between related Turkic languages, achieving +61% NER F1 improvement with retained source-language knowledge (Section 4.5).
4. We provide a quantitative **characterization of subword fragmentation** for Kyrgyz, Kazakh, and Uzbek (Section 3.6).
5. We release an open-source **SVD monitoring toolkit** for reproducible spectral analysis of LoRA training.

2 Related Work

2.1 LoRA and Parameter-Efficient Fine-Tuning

Low-Rank Adaptation (LoRA) (Hu et al., 2022) parameterizes weight updates as $\Delta W = BA$ where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with rank $r \ll \min(d, k)$, reducing trainable parameters by orders of magnitude. QLoRA (Dettmers et al., 2023) extends this approach by quantizing the base model to 4-bit precision, enabling fine-tuning

of multi-billion parameter models on consumer hardware. Subsequent work has explored rank allocation strategies (Zhang et al., 2023), training stability improvements (Lialin et al., 2023), and theoretical analyses of low-rank update expressiveness (Malladi et al., 2023).

2.2 Spectral Analysis of Neural Network Weights

The spectral properties of neural network weight matrices have been studied as indicators of training dynamics and generalization. Martin and Mahoney (2021) demonstrated that well-trained networks exhibit heavy-tailed singular value distributions, a property they termed “implicit self-regularization.” In the context of LoRA, Biderman et al. (2024) showed that LoRA adapters learn less and forget less than full fine-tuning, and identified spectral collapse as a potential diagnostic for training degradation. Jang et al. (2024) provided a more detailed analysis of spectral collapse dynamics, proposing that monitoring the spectral energy ratio can serve as an early warning system for adapter failure. However, both studies focused primarily on English-language settings with relatively large training corpora, leaving open the question of whether these findings generalize to low-resource, morphologically complex languages.

2.3 Low-Resource NLP for Turkic Languages

Turkic languages have received increasing attention in the NLP community, though they remain severely underrepresented relative to their speaker populations. Recent benchmarks such as TUMLU (Isbarov et al., 2024) provide standardized evaluation for Turkic language understanding. Work on Kazakh NLP has leveraged Wikipedia and web-crawled corpora (Yeshpanov et al., 2024), while Kyrgyz and Uzbek remain among the least-resourced languages with significant digital presence. The challenge of tokenizer mismatch for agglutinative languages has been documented (Rust et al., 2021; Ahia et al., 2023), with morphologically rich languages consistently suffering from higher subword fragmentation ratios that inflate sequence lengths and reduce the effective information density per token.

2.4 Catastrophic Forgetting in Fine-Tuning

Catastrophic forgetting (McCloskey and Cohen, 1989; Kirkpatrick et al., 2017)—the loss of previously learned knowledge when adapting to new tasks—is a persistent challenge in continual learning. While LoRA’s low-rank constraint was hypothesized to mitigate forgetting by limiting the magnitude of weight updates (Biderman et al., 2024), recent work has shown that aggressive fine-tuning can still induce significant forgetting, particularly for languages distant from the training data (Csaki et al., 2024). The relationship between spectral properties of adapter weights and the degree of catastrophic forgetting has not been systematically studied for multilingual settings.

3 Experimental Setup

3.1 Base Model

We use Gemma-2-9B (Gemma Team et al., 2024) as our base model, a 9.24 billion parameter decoder-only transformer with 42 layers. The model is quantized to 4-bit precision using the NF4 data type with double quantization (Dettmers et al., 2023), reducing memory requirements to approximately 5.1 billion effective parameters while preserving model quality. All experiments use bfloat16 compute precision.

3.2 Training Data

We compile monolingual corpora of approximately 150 MB for each of the three target languages:

- **Kazakh (KZ):** 61,879 records sourced from Kazakh Wikipedia and the sozkz-corpus, totaling 14.1M tokens (151.5 MB).
- **Kyrgyz (KY):** 17,360 records from a curated local corpus comprising literature, history, and encyclopedic texts, totaling 4.4M tokens (151.4 MB).
- **Uzbek (UZ):** 21,242 records from uz-books, Uzbek Wikipedia, and FineWeb-2 with Cyrillic script filtering, totaling 5.4M tokens (152.6 MB).

All texts are chunked into segments of 200–5,000 characters. A 10% held-out split is reserved for validation. The substantial difference in token counts between Kazakh (14.1M) and the other two

languages (4.4–5.4M) reflects the higher information density of Kazakh Wikipedia articles; all corpora are matched by raw byte size.

3.3 LoRA Configuration

We apply LoRA adapters to all attention and MLP projection matrices: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj` (7 modules per layer, 294 total across 42 layers). We set $\alpha = 2r$ in all experiments and use the paged AdamW optimizer with a cosine learning rate schedule and 5% warmup. Table 1 summarizes all experimental configurations.

3.4 SVD Monitoring Protocol

We perform singular value decomposition (SVD) of each `lora_B` weight matrix every 100 training steps, computing five spectral metrics:

Spectral Energy (SE) quantifies the dominance of the largest singular value:

$$SE = \frac{\sigma_1}{\sum_{i=1}^r \sigma_i} \quad (1)$$

Effective Rank (Roy and Vetterli, 2007) measures the number of significantly contributing singular values:

$$\text{EffRank} = \exp \left(- \sum_{i=1}^r p_i \log p_i \right), \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j} \quad (2)$$

Stable Rank provides a continuous relaxation of matrix rank:

$$\text{StableRank} = \frac{\|W\|_F^2}{\sigma_1^2} = \frac{\sum_i \sigma_i^2}{\sigma_1^2} \quad (3)$$

SVD Entropy measures the uniformity of the singular value distribution:

$$H_{\text{SVD}} = - \sum_{i=1}^r p_i \log_2 p_i \quad (4)$$

Frobenius Norm captures the overall magnitude of the weight matrix:

$$\|W\|_F = \sqrt{\sum_{i=1}^r \sigma_i^2} \quad (5)$$

We monitor both `lora_A` and `lora_B` Frobenius norms separately. The spectral collapse threshold is set at $SE = 0.7$ following Biderman et al. (2024).

ID	Experiment	Language	Rank	LR	Epochs	Trainable %	Init
E1	KZ Baseline	Kazakh	16	2×10^{-4}	3	1.05%	Random
E2	UZ Baseline	Uzbek	16	2×10^{-4}	3	1.05%	Random
E3	KY Baseline	Kyrgyz	16	2×10^{-4}	3	1.05%	Random
E4	KY Overfit	Kyrgyz	16	2×10^{-4}	10	1.05%	Random
E5	KY Collapse	Kyrgyz	64	5×10^{-4}	5	4.08%	Random
E6	KZ→KY Transfer	Kyrgyz	16	2×10^{-4}	3	1.05%	KZ ckpt
E7	Diverged	All	64	1×10^{-3}	3	4.08%	Random

Table 1: Summary of experimental configurations. All experiments use Gemma-2-9B with 4-bit quantization, $\alpha = 2r$, cosine LR schedule, and effective batch size 16. Experiment E7 diverged during training and is excluded from downstream evaluation.

3.5 Evaluation Protocol

We evaluate each adapter on three tasks:

Perplexity (PPL): Computed on held-out validation splits (500 samples, 113–128K tokens per language). We construct a *cross-lingual perplexity matrix* by evaluating each adapter on all three languages, providing a direct measure of catastrophic forgetting.

Named Entity Recognition (NER): Using the WikiANN dataset (Pan et al., 2017) with 3-shot prompting on 100 test samples per language. We report micro-averaged F1 across PER, ORG, and LOC entity types.

Question Answering (QA): Using TUMLU (Isbarov et al., 2024), a multiple-choice benchmark for Turkic language understanding, with 5-shot log-likelihood evaluation on 695–794 questions per language.

3.6 Subword Tokenization Analysis

We measure subword fragmentation as the average number of tokens per whitespace-delimited word for each language. As shown in Table 2, all three Turkic languages exhibit severe fragmentation at approximately $3.2\times$ the rate of English, reflecting the fundamental mismatch between the Gemma tokenizer (optimized for English and high-resource languages) and agglutinative Turkic morphology.

4 Results

4.1 Baseline Performance

Table 3 presents the primary evaluation metrics across all experiments. Several patterns emerge from the baseline comparisons.

The Kazakh baseline (E1) achieves the best target-language perplexity (2.73), likely reflecting both the larger effective token count (14.1M vs. 4.4–5.4M) and better representation in Gemma’s

Language	Tokens/Word	Ratio to EN
English	1.15	1.00×
Uzbek	3.68	3.20×
Kyrgyz	3.69	3.21×
Kazakh	3.74	3.25×

Table 2: Subword fragmentation ratios for the Gemma-2-9B tokenizer. Turkic languages require approximately $3.2\times$ more tokens per word than English, reflecting severe tokenizer mismatch for agglutinative morphology.

pretraining data. The Uzbek baseline (E2) achieves the highest NER F1 on its own language (0.287), though its perplexity (4.03) is higher than Kazakh. The Kyrgyz baseline (E3) shows the weakest overall performance (PPL 4.78, NER F1 0.157), consistent with Kyrgyz being the least resourced of the three languages.

The most striking result is the behavior of the $r=64$ configuration (E5): despite achieving a target-language perplexity of only 5.90 (a modest 23% increase over the baseline), it suffers *complete* NER failure with $F1 = 0$ across all three languages, driven by 100% parse failures. This indicates that the model has lost its instruction-following capability entirely, a form of functional collapse that perplexity alone fails to capture. We note that this NER failure may reflect loss of structured output formatting rather than loss of entity knowledge per se; however, the TUMLU evaluation, which uses log-likelihood scoring and does not require generation, corroborates the degradation: accuracy drops to 23.2% (near the 25% random baseline for 4-option multiple choice), confirming that the functional collapse extends beyond instruction-following to language understanding itself.

ID	Experiment	Target Lang	PPL (target)	NER F1 (KY)	NER F1 (KZ)	NER F1 (UZ)	TUMLU (target)	TUMLU (KZ)
E1	KZ Baseline	KZ	2.73	0.219	0.164	0.578	38.4%	38.4%
E2	UZ Baseline	UZ	4.03	0.138	0.167	0.287	35.4%	33.1%
E3	KY Baseline	KY	4.78	0.157	0.190	0.326	34.7%	30.2%
E4	KY Overfit (10ep)	KY	4.18	0.173	0.238	0.239	33.9%	27.8%
E5	KY $r=64$	KY	5.90	0.000	0.000	0.000	23.2%	22.2%
E6	KZ→KY	KY	4.73	0.253	0.130	0.348	33.7%	31.7%

Table 3: Main evaluation results across all experiments. PPL is computed on the target language validation set. NER F1 is micro-averaged across entity types using 3-shot WikiANN evaluation (100 samples). TUMLU accuracy is reported for the target language and Kazakh (as a forgetting diagnostic). The $r=64$ experiment (E5) shows complete NER failure (100/100 parse failures) despite only moderate PPL degradation.

Trained on	KY PPL	KZ PPL	UZ PPL
KZ Baseline	40.75	2.73	41.27
UZ Baseline	116.89	64.44	4.03
KY Baseline	4.78	47.58	56.85
KY Overfit	4.18	87.65	95.76
KY $r=64$	5.90	659.25	742.73
KZ→KY	4.73	23.49	124.19

Table 4: Cross-lingual perplexity matrix. Each row is an adapter; each column is the evaluation language. Bold indicates target-language PPL. The $r=64$ configuration shows catastrophic cross-lingual degradation ($> 600\times$ PPL increase), while KZ→KY transfer retains substantially more Kazakh knowledge (23.49 vs. 47.58).

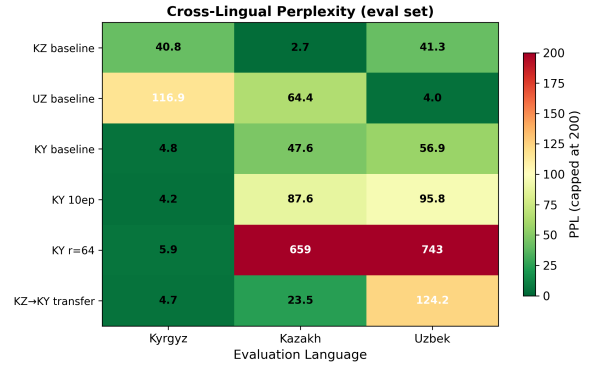


Figure 1: Cross-lingual perplexity heatmap. Darker cells indicate higher perplexity (more forgetting). The $r=64$ row is clearly pathological, while the KZ→KY transfer row shows the most balanced profile.

4.2 Cross-Lingual Degradation

The cross-lingual perplexity matrix (Table 4) reveals the extent of catastrophic forgetting induced by monolingual fine-tuning.

Several key observations emerge. First, the $r=64$ configuration (E5) induces catastrophic forgetting: Kazakh PPL rises to 659.25 and Uzbek to 742.73, representing increases of over $240\times$ and $180\times$ relative to the respective baselines. Second, the KZ→KY transfer (E6) retains substantially more Kazakh knowledge than direct Kyrgyz training: KZ PPL of 23.49 versus 47.58 for the KY baseline (a $2\times$ improvement). Third, extended training (10 epochs, E4) nearly doubles cross-lingual PPL relative to the 3-epoch baseline (87.65 vs. 47.58 for KZ; 95.76 vs. 56.85 for UZ), confirming that additional training steps systematically erode non-target-language knowledge. Fourth, the Uzbek baseline (E2) shows the highest asymmetric forgetting: KY PPL rises to 116.89, substantially worse than the reverse direction (UZ PPL under KY training is 56.85), possibly reflecting the greater script distance between Uzbek (historically Latin) and Kyrgyz (Cyrillic).

Figure 1 visualizes these patterns as a heatmap.

4.3 Spectral Dynamics: Refuting H1

Our central finding is that the spectral energy threshold of 0.7 proposed by Biderman et al. (2024) is **never reached** in any of our experiments, including those exhibiting severe functional degradation. Figure 2 shows the evolution of mean and maximum SE across training steps for representative experiments.

The KY baseline ($r=16$) shows a mild increase in maximum SE from approximately 0.44 to 0.59 over the course of training, but this remains well below the 0.7 threshold. More strikingly, the pathological $r=64$ configuration exhibits *decreasing* SE throughout training, from an initial value of approximately 0.21 to a stable range of 0.10–0.13. This counterintuitive pattern arises because SE is a *shape* metric that measures the relative dominance of the largest singular value: as training progresses, all singular values grow, but the growth is distributed across multiple directions, causing the relative contribution of σ_1 to decrease.

SVD Dynamics: Baseline vs Aggressive vs Transfer

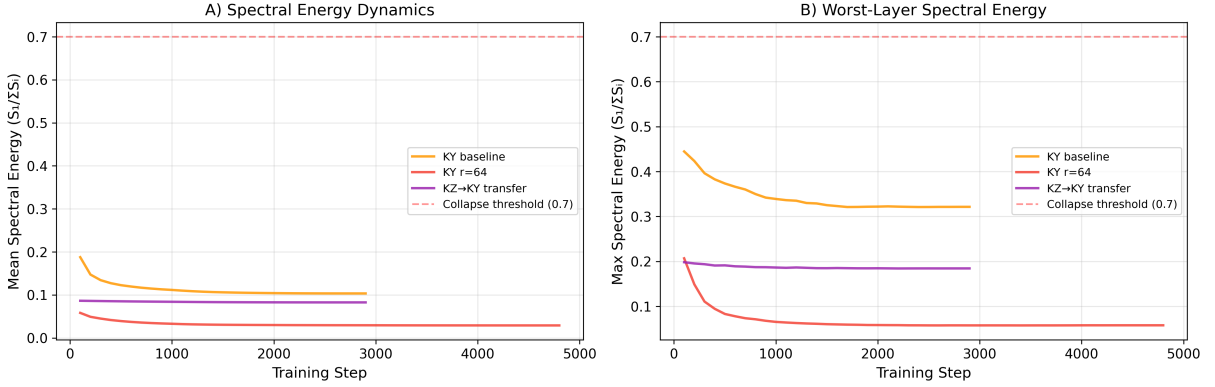


Figure 2: Spectral energy dynamics during training. Top: mean SE across all layers. Bottom: maximum SE across layers. No experiment approaches the proposed 0.7 threshold. The $r=64$ configuration shows *decreasing* SE despite suffering the worst functional degradation.

The effective rank tells a complementary story: for the $r=64$ configuration, effective rank *increases* from approximately 6 to 26, indicating that the adapter is utilizing more of its available capacity. Under standard interpretations, increasing effective rank signals healthy training dynamics. Yet this same configuration suffers complete NER failure and catastrophic cross-lingual forgetting.

Figure 3 shows the layer-wise SE distribution at the final checkpoint, confirming that no individual layer approaches the collapse threshold.

H1 is therefore refuted: spectral energy below the proposed threshold does *not* guarantee healthy training, and SE can decrease during training that leads to functional collapse. The SE metric fundamentally measures the *shape* of the singular value distribution while ignoring its *scale*—a critical limitation when diagnosing pathological training regimes.

4.4 Frobenius Norm: The Missing Diagnostic

If spectral energy fails as a diagnostic, what metric does reliably predict functional degradation? We find that the **Frobenius norm growth ratio** of the `lora_B` matrices provides a clear separation between healthy and pathological training regimes.

Table 5 and Figure 4 show a clear gradient: the KZ→KY transfer, which achieves the best downstream performance, exhibits only +13% Frobenius norm growth (from ~ 1.7 to ~ 1.9), beginning from the already-trained KZ checkpoint. The KY baseline shows moderate growth of approximately $8\times$ (from 0.22 to ~ 1.7). The pathologi-

Experiment	$\ B\ _F^{\text{init}}$	$\ B\ _F^{\text{final}}$	Growth
KZ→KY Transfer	~ 1.7	~ 1.9	$1.13\times$
KY Baseline $r=16$	0.22	~ 1.7	$\sim 8\times$
KY $r=64$	0.37–0.92	14–35	$15\text{--}38\times$
Diverged $r=64$	—	—	explodes

Table 5: Frobenius norm growth ratios for `lora_B` matrices. The transfer setup shows minimal growth (+13%), baseline shows moderate growth ($\sim 8\times$), while the pathological $r=64$ configuration shows extreme growth ($15\text{--}38\times$). The diverged experiment (E7) exhibits unbounded norm explosion.

cal $r=64$ configuration shows extreme growth of $15\text{--}38\times$ (from 0.37–0.92 to 14–35, varying across layers). The diverged experiment (E7, $r=64$, $\text{lr} = 1 \times 10^{-3}$) shows unbounded norm explosion before training terminates.

The Frobenius norm growth ratio correlates strongly with cross-lingual PPL degradation (Figure 5). This relationship is intuitive: the Frobenius norm is a *scale-aware* metric that captures the total magnitude of weight updates. Large weight perturbations are more likely to disrupt the pre-trained model’s representations, leading to catastrophic forgetting. Unlike SE, which normalizes out scale information, $\|B\|_F$ directly reflects how far the adapter has moved from its initialization.

4.5 Cross-Lingual Transfer

The KZ→KY transfer experiment (E6) initializes LoRA adapters from the final Kazakh checkpoint (E1) before continuing training on Kyrgyz data. This simple sequential transfer yields substantial improvements across multiple metrics (Table 6).

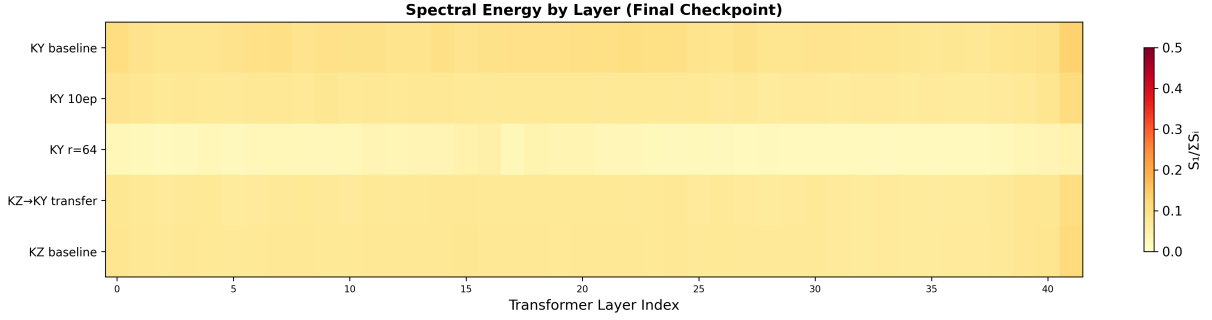


Figure 3: Layer-wise spectral energy at the final training checkpoint. Each row is an experiment; each column is a transformer layer. No layer in any experiment exceeds the 0.7 threshold.

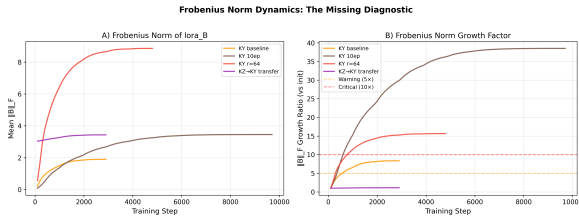


Figure 4: Frobenius norm dynamics of `lora_B` matrices during training. The $r=64$ configuration shows dramatically larger norm growth compared to $r=16$ baselines. The transfer experiment begins at the KZ baseline’s final norm and shows minimal additional growth.

The most dramatic improvement is in Kyrgyz NER F1: 0.253 versus 0.157, a relative improvement of +61%. Target-language perplexity is essentially identical (4.73 vs. 4.78), indicating that the transfer does not sacrifice language modeling quality. Kazakh knowledge is substantially better retained: KZ PPL of 23.49 versus 47.58 (a $2\times$ improvement), and KZ TUMLU accuracy improves from 30.2% to 31.7%.

Interestingly, even Uzbek NER F1 improves slightly under transfer (0.348 vs. 0.326), suggesting some positive transfer to the third Turkic language. However, Uzbek PPL degrades more substantially (124.19 vs. 56.85), indicating that the transfer benefit is primarily concentrated between Kazakh and Kyrgyz—both Kipchak-branch Turkic languages sharing Cyrillic script—and does not fully extend to the more distant Uzbek (Karluk branch).

H2 is partially confirmed: cross-lingual transfer improves downstream task performance and retains source-language knowledge, but the mechanism appears to be *better initialization* rather than *slower degradation*. The transfer adapter begins with an already-trained weight configuration (Frobenius norm growth of only +13% compared

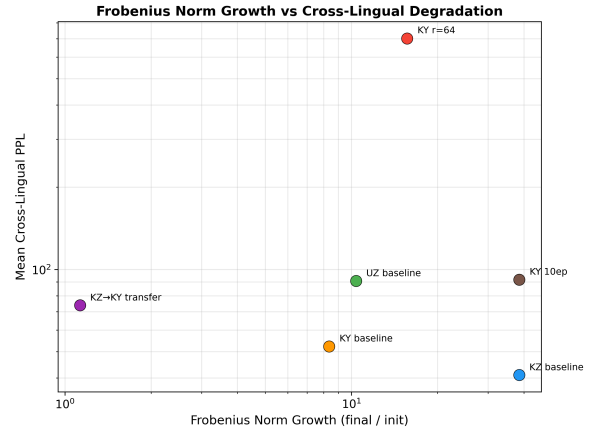


Figure 5: Frobenius norm growth ratio versus cross-lingual perplexity degradation. Larger norm growth is associated with more severe catastrophic forgetting, providing a practical early-stopping criterion.

to $\sim 8\times$ for baseline), requiring minimal additional adaptation to accommodate Kyrgyz data.

4.6 The Overfit Paradox

The 10-epoch Kyrgyz experiment (E4) reveals a paradoxical pattern (Figure 6): extended training *improves* target-language perplexity (4.18 vs. 4.78 for the 3-epoch baseline) despite clear indicators of overfitting (final train loss 0.165 vs. 0.815 for the baseline, indicating severe memorization).

However, this improvement comes at a steep cost in cross-lingual knowledge. Kazakh PPL nearly doubles (87.65 vs. 47.58), Uzbek PPL increases by 68% (95.76 vs. 56.85), and TUMLU accuracy on Kazakh drops from 30.2% to 27.8%. The spectral profile remains healthy throughout all 10 epochs ($SE < 0.5$), further confirming that SE is not a reliable indicator of training pathology.

This pattern suggests a regime of “useful overfitting” for the target language: the model memorizes patterns specific to Kyrgyz text more

Metric	KY Baseline	KZ→KY
PPL (KY)	4.78	4.73
PPL (KZ)	47.58	23.49
PPL (UZ)	56.85	124.19
NER F1 (KY)	0.157	0.253
NER F1 (KZ)	0.190	0.130
NER F1 (UZ)	0.326	0.348
TUMLU (KY)	34.7%	33.7%
TUMLU (KZ)	30.2%	31.7%
$\ B\ _F$ growth	$\sim 8\times$	$1.13\times$

Table 6: Comparison of direct Kyrgyz training (E3) and Kazakh→Kyrgyz transfer (E6). Transfer improves KY NER F1 by +61% and retains KZ knowledge (PPL 23.49 vs. 47.58). However, UZ PPL increases (124.19 vs. 56.85), suggesting the transfer benefit is primarily between the two closely related Cyrillic-script Kipchak languages.

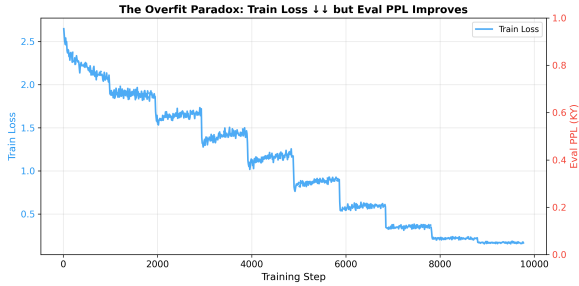


Figure 6: The overfit paradox: 10 epochs of Kyrgyz training (E4) yields better target-language PPL (4.18 vs. 4.78) despite increasing cross-lingual degradation. Train loss continues to decrease while cross-lingual PPL rises monotonically.

deeply, yielding better held-out perplexity on in-distribution data, but at the expense of broader multilingual competence. For practitioners focused on a single target language, extended training may be acceptable; for those seeking to maintain multilingual capability, it represents a significant regression.

5 Analysis and Discussion

5.1 Why Spectral Energy Fails as a Diagnostic

The failure of the SE threshold can be understood through the mathematics of LoRA initialization. Standard LoRA initializes A with Kaiming uniform and B with zeros. After the first gradient update, B acquires small random-scale singular values with roughly uniform magnitudes, yielding $SE \approx 1/r$. As training progresses, all singular values grow, but the growth is typically distributed across multiple directions, causing SE to remain

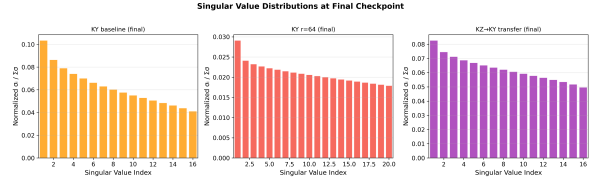


Figure 7: Singular value distributions at the final checkpoint. The $r=64$ configuration (right) shows a well-distributed spectrum (low SE) with dramatically larger absolute magnitudes compared to the $r=16$ baseline (left), explaining why SE fails to detect pathological training.

stable or even *decrease*.

For higher-rank configurations ($r=64$), the SE denominator includes more singular values, mechanically driving SE to lower values ($1/64 \approx 0.016$ at initialization versus $1/16 = 0.0625$ for $r=16$). This reveals SE as a *rank-biased metric*: since $SE \approx 1/r$ at initialization, higher-rank adapters *appear* spectrally healthier by definition, creating a dangerous trap for practitioners who monitor SE across configurations with different ranks. A seemingly healthy low-SE value for $r=64$ may actually correspond to a more pathological state than a higher SE for $r=16$.

The fundamental limitation is that SE is a *shape* metric, invariant to rescaling of the singular value spectrum. A weight matrix with singular values $[100, 50, 25]$ and one with $[0.001, 0.0005, 0.00025]$ have identical SE, despite representing vastly different magnitudes of weight perturbation. In the low-resource setting, where models must substantially restructure their representations to accommodate poorly-covered languages, the *scale* of these perturbations matters critically.

Figure 7 visualizes the singular value distributions for representative experiments, illustrating how the $r=64$ configuration develops a well-distributed but extremely large-magnitude spectrum that SE fails to flag.

5.2 Frobenius Norm as a Scale-Aware Alternative

The Frobenius norm, $\|W\|_F = \sqrt{\sum_i \sigma_i^2}$, naturally captures the scale of weight perturbations that SE discards. Its squared value equals the sum of squared singular values, providing a direct measure of the total energy in the weight update. The Frobenius norm growth ratio (final/initial) thus quantifies how much the adapter has perturbed the

pretrained weights.

Our experiments suggest the following practical thresholds for $\|B\|_F$ growth monitoring:

- $< 2\times$: **Healthy** training; minimal forgetting (e.g., transfer setup: $1.13\times$).
- $2\text{--}10\times$: **Normal** training; moderate forgetting expected (e.g., baseline: $\sim 8\times$).
- $> 10\times$: **Warning**; significant forgetting likely (e.g., $r=64$: $15\text{--}38\times$).
- Unbounded growth: **Training failure**; divergence imminent (e.g., E7).

We emphasize that these thresholds refer to *relative growth* (ratio of final to initial Frobenius norm) rather than absolute values. Absolute norms depend on the initialization scheme: standard LoRA (zeros for B , Kaiming for A), PiSSA (Meng et al., 2024), or transfer initialization all produce different baseline magnitudes. The relative growth ratio normalizes this dependency, making it a more portable diagnostic. Nonetheless, these thresholds are preliminary and derived from a single model architecture; we present them as guidelines for further investigation rather than universal constants.

5.3 Role of Subword Fragmentation

The $3.2\times$ subword fragmentation ratio observed for all three Turkic languages has several compounding effects on LoRA adaptation:

Sequence length inflation: A Kyrgyz sentence of 50 words requires approximately $50 \times 3.69 = 185$ tokens, versus $50 \times 1.15 = 58$ tokens for the English equivalent. With a maximum sequence length of 256 tokens, each training example contains substantially less textual content, reducing the information density of each gradient update.

Morphological fragmentation: Agglutinative suffixes that carry grammatical meaning are split across multiple subwords, forcing the model to reconstruct morphological relationships across token boundaries. For instance, a single Kazakh word form may be split into 4–5 subwords, with the root, derivational suffixes, and inflectional endings each occupying separate tokens.

Potential link to norm growth: The model must make larger modifications to its internal representations to accommodate the fragmented Turkic input, potentially explaining the faster Frobenius norm growth observed in our experiments.

However, without an English-language control experiment with matched data size, this connection remains speculative and we present it as a direction for future work rather than a confirmed finding.

5.4 Practical Recommendations

Based on our findings, we offer the following recommendations for practitioners fine-tuning LLMs on low-resource, morphologically complex languages:

1. **Monitor Frobenius norm growth**, not spectral energy. Set warning thresholds at $5\times$ growth and stopping thresholds at $10\times$ growth of $\|B\|_F$.
2. **Prefer lower ranks** ($r=16$) for small corpora (~ 150 MB). Higher ranks ($r=64$) lead to functional collapse in our setting, even when standard spectral metrics appear healthy.
3. **Use related-language transfer** when available. Initializing from a related-language checkpoint provides better weight initialization and substantially improves downstream task performance with minimal additional Frobenius norm growth.
4. **Apply cross-lingual perplexity monitoring** as a forgetting diagnostic. Target-language PPL alone is insufficient; catastrophic forgetting manifests primarily in non-target languages.
5. **Consider early stopping by cross-lingual PPL**, not training loss. Extended training may improve target-language metrics while severely degrading multilingual capability.

6 Limitations

Our study has several important limitations. First, all experiments use a single base model (Gemma-2-9B); results may differ for other architectures such as Llama 3 or Mistral, particularly regarding the relationship between rank, norm growth, and functional collapse. Second, the absence of an English-language control experiment with matched data size means we cannot isolate the effect of Turkic morphology on Frobenius norm growth from other confounding factors such as data distribution, corpus size, and tokenizer coverage. Third, NER evaluation relies

on few-shot prompting rather than a fine-tuned NER model, introducing noise from instruction-following variation; the complete NER failure for $r=64$ (100/100 parse failures) may partially reflect loss of instruction-following capability rather than loss of named entity knowledge per se. Fourth, we examine only three of the approximately 35 Turkic languages, excluding Turkish (the highest-resource Turkic language), limiting the generalizability of our cross-lingual transfer findings. Fifth, the fixed corpus size of approximately 150 MB per language precludes scaling analyses that might reveal different dynamics at larger data volumes. Sixth, our SVD monitoring focuses on `lora_B` matrices; `lora_A` dynamics may reveal complementary patterns. Seventh, 4-bit quantization introduces approximation errors that may influence spectral properties in ways not captured by our analysis. Finally, our Frobenius norm thresholds are derived from a single experimental setting and should be treated as preliminary guidelines pending validation across diverse models and languages.

7 Conclusion

We have presented a systematic study of the spectral dynamics of LoRA adapters during fine-tuning on three low-resource Turkic languages. Our experiments yield three principal findings.

First, the spectral energy threshold hypothesis is **refuted** for our setting: no experiment exceeds SE of 0.59 (well below the proposed 0.7 threshold), yet configurations with $r=64$ suffer complete functional collapse (NER F1 = 0, cross-lingual PPL > 600). Spectral energy decreases during training as effective rank grows, rendering it an unreliable diagnostic for training pathology.

Second, **Frobenius norm growth ratio** emerges as a more reliable indicator of functional degradation. Pathological configurations exhibit $15\text{--}38\times$ norm growth versus $1.13\times$ for the best-performing transfer setup, with clear correlation to cross-lingual perplexity degradation. We propose preliminary monitoring thresholds ($> 5\times$ warning, $> 10\times$ stop), noting that these are specific to our experimental setting (Gemma-2-9B with 4-bit NF4 quantization and standard LoRA initialization) and require validation across architectures before general adoption.

Third, **sequential cross-lingual transfer** between related Turkic languages is highly effective:

Kazakh→Kyrgyz transfer improves NER F1 by +61% (0.253 vs. 0.157) while retaining Kazakh knowledge (PPL 23.49 vs. 47.58), with minimal Frobenius norm growth (+13%).

These findings have immediate practical implications for the growing community of researchers and practitioners adapting LLMs to low-resource languages. Future work should validate our Frobenius norm thresholds across diverse model architectures, extend the analysis to additional Turkic languages including Turkish, investigate Frobenius norm regularization as a training objective, and explore adaptive rank selection strategies informed by real-time spectral monitoring. An English-language control experiment with matched data size would help disentangle the effects of Turkic morphological complexity from general low-resource dynamics. Additionally, log-likelihood-based NER evaluation could clarify whether the $r=64$ configuration loses named entity knowledge entirely or merely loses the instruction-following capability needed for structured output.

Ethics Statement

This work involves fine-tuning language models on publicly available text corpora from three Turkic languages. While we aim to improve NLP capabilities for underserved language communities, we acknowledge that language models can amplify biases present in training data. Our training corpora, drawn from Wikipedia, literary texts, and web sources, may contain cultural biases or factual inaccuracies. We do not release fine-tuned model weights, only the monitoring toolkit and training code.

Acknowledgments

We thank the creators of the Kazakh, Kyrgyz, and Uzbek corpora used in this study, as well as the developers of the TUMLU benchmark for providing standardized Turkic language evaluation resources. All experiments were conducted on personal hardware without external funding.

References

Orevaoghene Ahia, Julia Kreutzer, and Sara Hooker. 2023. Do all languages cost the same? tokenization in the era of commercial language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.

- Dan Biderman, Jose Gonzalez Portes, Alon Spirber, Keiran Paster, Elias Frantar, and J Zico Kolter. 2024. LoRA learns less and forgets less. In *Transactions on Machine Learning Research*.
- Zoltan Csaki, Bo Peng, Douglas Steiner, and 1 others. 2024. Sambalingo: Teaching large language models new languages. In *arXiv preprint arXiv:2404.05829*.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient finetuning of quantized language models. In *Advances in Neural Information Processing Systems*.
- Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, and 1 others. 2024. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.
- Maral Isbarov and 1 others. 2024. TUMLU: Turkic multi-task language understanding benchmark. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.
- Uijeong Jang, Jason D Lee, and Ernest K Ryu. 2024. Lora training in the NTK regime has no spurious local minima. In *International Conference on Machine Learning*.
- James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Oriol Vinyals, Guillaume Desjardins, Andrei A Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, and 1 others. 2017. Overcoming catastrophic forgetting in neural networks. In *Proceedings of the National Academy of Sciences*, volume 114, pages 3521–3526.
- Vladislav Lialin, Namrata Deshpande, and Anna Rumshisky. 2023. ReLoRA: High-rank training through low-rank updates. In *NeurIPS 2023 Workshop on Efficient Natural Language and Speech Processing*.
- Sadhika Malladi, Tianyu Gao, Eshaan Nichani, Alex Damian, Jason D Lee, Danqi Chen, and Sanjeev Arora. 2023. A kernel-based view of language model fine-tuning. In *International Conference on Machine Learning*.
- Charles H Martin and Michael W Mahoney. 2021. Implicit self-regularization in deep neural networks: Evidence from random matrix theory and implications for training. In *Journal of Machine Learning Research*, volume 22, pages 1–73.
- Michael McCloskey and Neal J Cohen. 1989. Catastrophic interference in connectionist networks: The sequential learning problem. *Psychology of Learning and Motivation*, 24:109–165.
- Fanxu Meng, Zhaohui Wang, and Muhan Zhang. 2024. PiSSA: Principal singular values and singular vectors adaptation of large language models. *Advances in Neural Information Processing Systems*.
- Xiaoman Pan, Boliang Zhang, Jonathan May, Joel Nothman, Kevin Knight, and Heng Ji. 2017. Cross-lingual name tagging and linking for 282 languages. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*.
- Olivier Roy and Martin Vetterli. 2007. The effective rank: A measure of effective dimensionality. *European Signal Processing Conference*, pages 606–610.
- Phillip Rust, Jonas Pfeiffer, Ivan Vulic, Sebastian Ruder, and Iryna Gurevych. 2021. How good is your tokenizer? on the monolingual performance of multilingual language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*.
- Ruslan Yeshpanov, Yerbolat Khassanov, and Huseyin Atakan Varol. 2024. Kazllm: A study of large language models for kazakh language processing. *arXiv preprint arXiv:2407.15454*.
- Qingru Zhang, Minshuo Chen, Alexander Bukharin, Pengcheng Kaez, Niao He, and Tuo Huang. 2023. AdaLoRA: Adaptive budget allocation for parameter-efficient fine-tuning. In *International Conference on Learning Representations*.

A Detailed Training Configurations

All experiments were conducted on a single NVIDIA GeForce RTX 5080 (16 GB) using PyTorch 2.10, Transformers 5.2, and PEFT 0.18. Table 7 provides the full training statistics.

B Effective Rank Dynamics

Figure 8 shows the effective rank dynamics across experiments. The $r=64$ configuration shows increasing effective rank (from ~ 6 to ~ 26), indicating growing parameter utilization. The diverged experiment (E7) is the only configuration where effective rank *decreases*, consistent with true spectral collapse.

C Per-Type NER Results

Table 8 breaks down NER performance by entity type for the Kyrgyz evaluation.

D TUMLU Accuracy by Language

Table 9 presents TUMLU accuracy scores across all languages and experiments.

ID	Experiment	Train Loss	Eval Loss	Eval PPL	Tokens	Time (min)
E1	KZ Baseline	0.275	1.259	3.52	14.1M	477
E2	UZ Baseline	1.193	1.822	6.18	5.4M	341
E3	KY Baseline	0.815	1.920	6.82	4.4M	183
E4	KY Overfit	1.050	1.963	7.12	4.4M	1193
E5	KY $r=64$	0.274	2.164	8.71	4.4M	157
E6	KZ→KY	1.809	1.932	6.90	4.4M	355

Table 7: Full training statistics for all experiments. Eval loss and PPL are computed on the validation split during training (distinct from the post-hoc evaluation in Table 3). The higher train loss for E6 (KZ→KY) reflects the adapter starting from KZ-optimized weights that initially perform poorly on KY data.

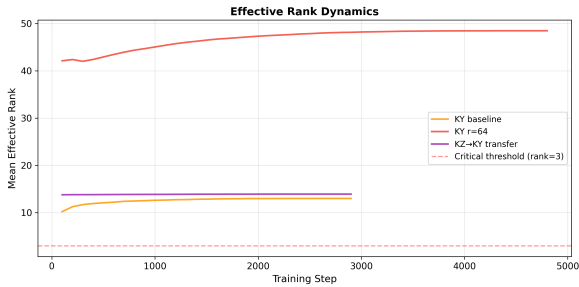


Figure 8: Effective rank dynamics during training. The $r=64$ configuration shows increasing effective rank despite functional degradation, further evidence that standard spectral metrics can be misleading.

Experiment	PER F1	ORG F1	LOC F1
KZ Baseline (on KY)	0.236	0.162	0.229
UZ Baseline (on KY)	0.095	0.150	0.209
KY Baseline	0.101	0.160	0.266
KY Overfit	0.224	0.255	0.119
KY $r=64$	0.000	0.000	0.000
KZ→KY	0.344	0.202	0.214

Table 8: Per-type NER F1 scores on Kyrgyz WikiANN. The KZ→KY transfer achieves the highest PER F1 (0.344), substantially outperforming the KY baseline (0.101). The $r=64$ configuration produces zero scores across all types due to 100% parse failures.

Experiment	KY	KZ	UZ
KZ Baseline	35.7%	38.4%	32.8%
UZ Baseline	31.9%	33.1%	35.4%
KY Baseline	34.7%	30.2%	29.6%
KY Overfit	33.9%	27.8%	31.5%
KY $r=64$	23.2%	22.2%	22.9%
KZ→KY	33.7%	31.7%	30.8%

Table 9: TUMLU accuracy across all experiments and languages. The KZ baseline achieves the highest accuracy on its target language (38.4%). The $r=64$ configuration drops to near-random levels ($\sim 23\%$ on 4-option multiple choice) across all languages.